

Plastics

Many artificial polymers are plastics. A plastic is an incredibly useful material that can be molded into any shape while hot, becoming solid when cool. It can also be pulled into thin films and used as a protective coating. Plastic is made from monomers derived from crude oil.

▽ Common plastics

Several plastics have become very familiar over the last few decades, because they have a huge range of applications.

Polymer	Monomer	Properties of polymer
polythene (polyethylene)	ethene	makes flexible plastics; is used in packaging and to insulate electrical wires
polystyrene	styrene	used to make Styrofoam; is also added to other polymers to make them waterproof
PVC (polyvinyl chloride)	chloroethene (vinyl chloride)	makes very tough plastics; is not damaged by strong chemicals; is a good insulator
teflon (polytetrafluoroethylene)	tetrafluoroethylene	a very slippery substance that is used on nonstick pans

Properties of plastics

It is easier to shape plastic polymers while they are warm or melted into a liquid. There are two main types of plastic. Thermoplastics can be molded, melted, and reshaped repeatedly. Thermosets can only be molded once; after they have set, they will burn without melting if reheated.

▷ Polymer properties

The properties of a polymer result from the shape of the monomers. Thermosets form crosslinks when solid, which make the polymer into a rigid lattice.

REAL WORLD

Rubber

The bark of rubber trees produces an oily liquid called latex that contains the compound isoprene. Adding an acid makes the isoprene in the liquid polymerize into solid rubber, which can be made into sheets or molded before it dries out.

